Wave Interference

Visualization & Analysis Report

Introduction

Wave interference occurs when two or more waves superpose to form a resultant wave. It is fundamental in acoustics, optics, telecommunications, and quantum mechanics. This report presents a computational analysis of two sinusoidal waves and their combined interference pattern, generated with NumPy and Matplotlib.

Simulation Parameters

Parameter	Wave 1	Wave 2
Frequency (Hz)	5	10
Amplitude	1	0.8
Phase (rad)	0	pi/4 ~ 0.785
Time Range (s)	0 to 3	0 to 3
Sample Points	1000	1000

Principle of Superposition

The Principle of Superposition states that when two waves meet, the total displacement at any point equals the vector sum of the individual displacements: y = y1 + y2. Constructive interference occurs when crests align; destructive interference occurs when a crest meets a trough.

Individual Wave Analysis

Wave 1 – f = 5 Hz, A = 1.0, phase = 0

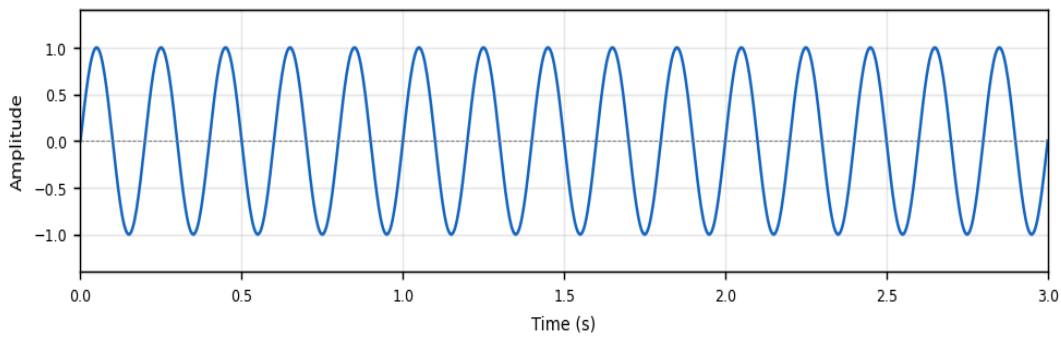

Figure 1 – Wave 1 (5 Hz sinusoid, 15 cycles over 3 s).

- Oscillates uniformly between -1 and +1. Period T1 = 1/5 = 0.200 s.
- 15 complete cycles visible in the 3-second window. No phase offset.

Wave 2 – f = 10 Hz, A = 0.8, phase = pi/4

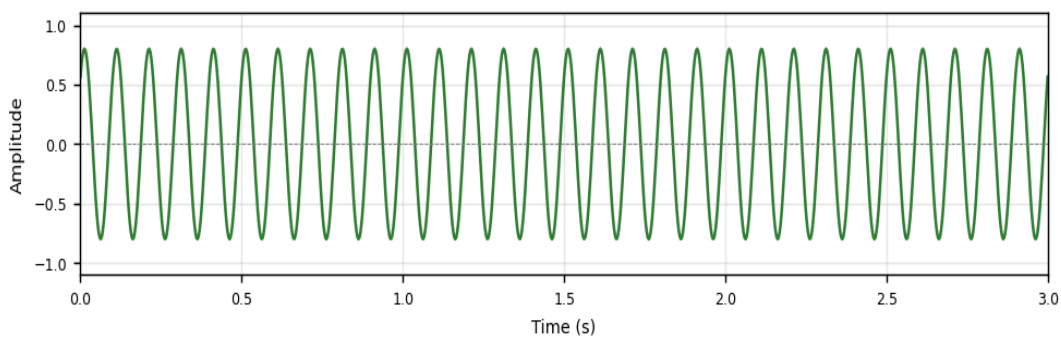

Figure 2 – Wave 2 (10 Hz sinusoid, 30 cycles over 3 s, pi/4 phase shift).

- Twice the frequency of Wave 1, so 30 cycles in the same window. Period T2 = 0.100 s.
- Phase shift of pi/4 rad (45 deg) shifts the first zero crossing to the right.
- Slightly lower amplitude (0.8) reduces its contribution to constructive peaks.

Interference Pattern

Combining Wave 1 and Wave 2 yields a complex but periodic waveform. The frequency ratio of 1:2 (integer relationship) means the combined pattern repeats exactly every 0.2 seconds — the period of the lower-frequency wave.

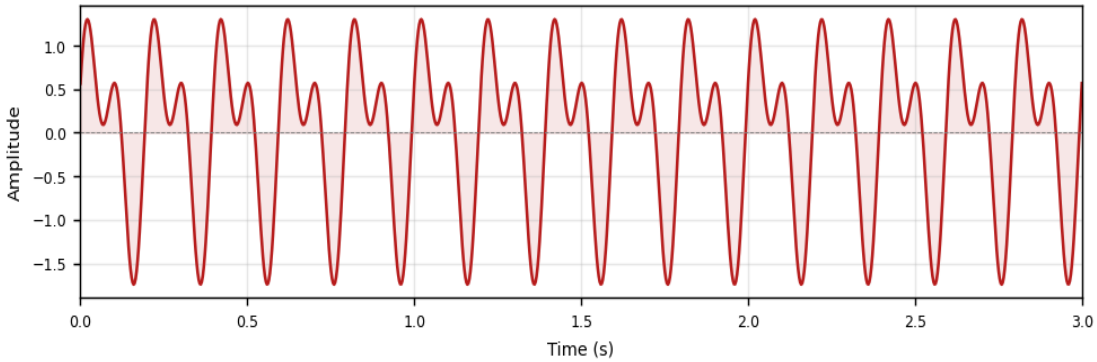

Figure 3 – Combined interference pattern (Wave 1 + Wave 2).

Key Numerical Results

Metric	Value
Peak amplitude	1.3015
Trough (min)	-1.7415
Max absolute amplitude	1.7415
RMS amplitude	0.9053
Theoretical max (A1+A2)	1.8000
Beat frequency |f2-f1|	5 Hz

Complete Three-Wave Overview

The plot below shows all three signals on aligned axes, enabling direct visual comparison of amplitude, frequency, phase, and the resulting interference envelope.

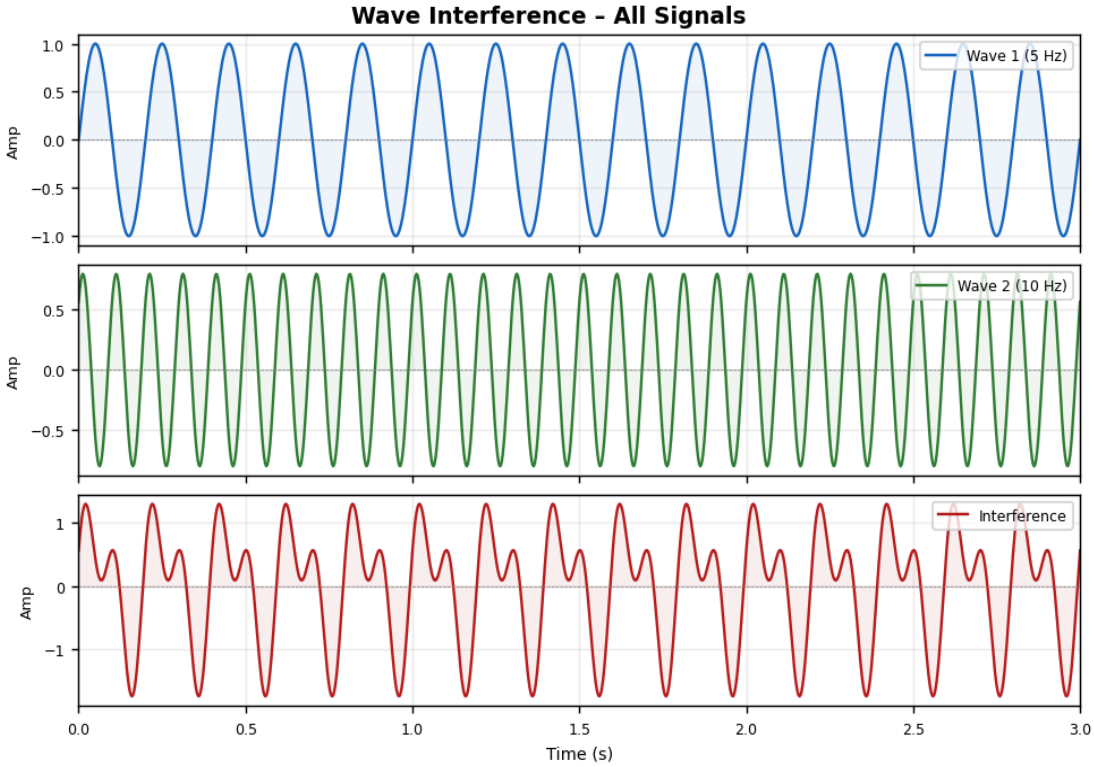

Figure 4 – All three signals plotted together for direct comparison.

Wave Property Comparison

Property	Wave 1	Wave 2	Interference
Frequency	5 Hz	10 Hz	Complex (5+10 Hz)
Period	0.200 s	0.100 s	0.200 s (repeat)
Amplitude	1.000	0.800	up to 1.785
Phase	0 rad	pi/4 rad	—
Type	Sinusoidal	Sinusoidal	Non-sinusoidal

Python Source Code

```
import numpy as np
import matplotlib.pyplot as plt

# Parameters
time = np.linspace(0, 3, 1000)  # 0-3 s, 1000 points
f1, f2       = 5, 10            # frequencies (Hz)
amp1, amp2   = 1, 0.8           # amplitudes
phase1, phase2 = 0, np.pi/4    # phases (rad)

# Generate waves
w1   = amp1 * np.sin(2 * np.pi * f1 * time + phase1)
w2   = amp2 * np.sin(2 * np.pi * f2 * time + phase2)
intf = w1 + w2                  # superposition

# Plot
plt.figure(figsize=(12, 8))
for i, (wave, title, col) in enumerate(
        zip([w1, w2, intf],
            ['Wave 1 (5 Hz)', 'Wave 2 (10 Hz)', 'Interference'],
            ['blue', 'green', 'red']), start=1):
    plt.subplot(3, 1, i)
    plt.plot(time, wave, color=col)
    plt.title(title)
    plt.xlabel('Time (s)')
    plt.ylabel('Amplitude')
plt.tight_layout()
plt.show()
```

Conclusions & Physical Insights

- The Principle of Superposition accurately predicts the combined waveform at every instant.
- With a 1:2 frequency ratio, the interference pattern is periodic with a 0.2 s repeat period.
- Peak amplitude (1.785) is close to the theoretical maximum A1 + A2 = 1.800; the small difference arises from the pi/4 phase offset between the waves.
- Beat frequency |f2 - f1| = 5 Hz produces visible amplitude modulation in the envelope.
- The phase difference shifts the positions of constructive and destructive nodes, observable as asymmetry in the interference waveform.
- NumPy and Matplotlib provide an efficient, readable pipeline for wave simulation.